

EMBEDDED SYSTEM TAXK -1

INTRODUCTION:

What is an embedded system?

[Embedded system](#) is a computational system that is developed based on an integration of both hardware and software in order to perform a given task. It can be said as a dedicated computer system has been developed for some particular reason. But it is not our traditional computer system or general-purpose computers, these are the Embedded systems that may work independently or attached to a larger system to work on a few specific functions. These embedded systems can work without human intervention or with little human intervention.

Components of Embedded Systems

1. Hardware 2. Software 3. Firmware

Examples of Embedded Systems

- Digital watches
- Washing Machine
- Toys
- Televisions
- Digital phones
- Laser Printer
- Cameras
- Industrial machines
- Electronic Calculators
- Automobiles
- Medical Equipment

Characteristics of an Embedded System

- **Performs specific task:** Embedded systems perform some specific function or tasks.
- **Low Cost:** The price of an embedded system is not so expensive.
- **Time Specific:** It performs the tasks within a certain time frame.
- **Low Power:** Embedded Systems don't require much power to operate.
- **High Efficiency:** The efficiency level of embedded systems is so high.
- **Minimal User interface:** These systems require less user interface and are easy to use.
- **Less Human intervention:** Embedded systems require no human intervention or very less human intervention.
- **Highly Stable:** Embedded systems do not change frequently mostly fixed maintaining stability.
- **High Reliability:** Embedded systems are reliable they perform tasks consistently well.
- **Use microprocessors or microcontrollers:** Embedded systems use [microprocessors](#) or [microcontrollers](#) to design and use limited memory.
- **Manufacturable:** The majority of embedded systems are compact and affordable to manufacture. They are based on the size and low complexity of the hardware.

Real word application

1 Smart home

Embedded systems make homes more convenient, secure, and energy-efficient

2 Automotive embedded system

Modern vehicles use many embedded systems for safety and performance.

3 Medical embedded system

Embedded systems are widely used in medical equipment for monitoring and treatment.

4 Robot

Robots use embedded systems for sensing, processing, and controlling movements

Difference between microcontroller and microprocessor

Feature	Microcontroller (MCU)	Microprocessor (MPU)
Architecture	Integrates CPU, RAM, ROM, and I/O all on a single chip.	Only the CPU is on the chip; requires external memory and I/O chips.
Primary Focus	Embedded control and specific, predefined tasks.	General-purpose, heavy, and complex computing.
Processing Power	Lower (typically in the MHz range).	Very high (typically in the GHz range).
Operating System	Usually runs straight firmware without an OS (or an RTOS).	Requires a full operating system (e.g., Windows, Linux, macOS).
Power Consumption	Low; highly optimized for long battery life.	High; needs a stable, continuous power source.
Cost	Very inexpensive.	Significantly more expensive.
Common Examples	Arduino, STM32, PIC, ESP32.	Intel Core, AMD Ryzen, Apple Silicon.

Component Study:

o Research and explain the role of Microcontrollers

A **microcontroller** is a compact integrated circuit (IC) that contains a **CPU, memory (RAM & Flash), and input/output (I/O) ports** on a single chip. It is used to control and automate electronic devices.

Ardunio

Arduino is a popular microcontroller platform used for learning, prototyping, and embedded system development.

Features

- Easy to program using the Arduino IDE
- Large community support

- Supports digital and analog input/output
- Compatible with many sensors and modules

Applications

- Home automation
- Robotics
- Smart lighting
- Weather monitoring
- IoT prototypes

ESP32

ESP32 is a powerful microcontroller designed for **Internet of Things (IoT)** applications. It has built-in Wi-Fi and Bluetooth.

Features

- Dual-core processor
- Built-in Wi-Fi and Bluetooth
- Low power consumption
- Multiple GPIO, ADC, DAC, PWM interfaces

Applications

- Smart home devices
- Wireless sensor networks
- Smart agriculture
- Wearable devices
- Remote monitoring system

Raspberry Pi Pico

Raspberry Pi Pico is a microcontroller board based on the RP2040 chip. It is designed for embedded systems, automation, and educational projects.

Features

- Dual-core ARM Cortex-M0+ processor
- 26 GPIO pins
- Low power consumption
- Supports C/C++ and MicroPython programming

Applications

- Robotics
- Industrial automation
- Sensor-based projects
- Home automation

o Basic sensors

Temperature Sensor

A **temperature sensor** is an electronic device that measures the temperature of its surroundings and converts it into an electrical signal that can be processed by a microcontroller. It is commonly used in embedded systems to monitor and control temperature automatically. Temperature sensors are widely used in air conditioners, refrigerators, weather stations, medical thermometers, and industrial automation systems. They provide accurate temperature readings, consume low power, and help improve the efficiency and safety of electronic devices.

Motion Sensor

A **motion sensor** is a device that detects the movement of people or objects within its sensing range. It works using technologies such as Passive Infrared (PIR), ultrasonic, or microwave sensing and sends a signal to the microcontroller whenever motion is detected. Motion sensors are commonly used in security alarm systems, automatic doors, smart lighting, CCTV cameras, and home automation systems. They improve security, reduce energy consumption, and provide quick and reliable detection of movement.

Light Sensor

A **light sensor** is an electronic device that measures the intensity of light in the surrounding environment. It usually uses a Light Dependent Resistor (LDR) or a photodiode, whose electrical properties change according to the amount of light falling on it. The sensor sends this information to a microcontroller, which can automatically control connected devices. Light sensors are used in automatic street lights, mobile phone brightness control, solar tracking systems, digital cameras, and smart home lighting. They help save energy by automatically adjusting lighting based on ambient light conditions.

MINI PROJECT –

Smart Lighting System (LED ON/OFF Based on Sensor Input)

A **Smart Lighting System** is an embedded system that automatically controls the lighting based on input from a sensor. It helps save electricity, increases convenience, and improves energy efficiency. The system commonly uses a **light sensor (LDR)** or a **motion sensor (PIR)** connected to a microcontroller such as **Arduino**.

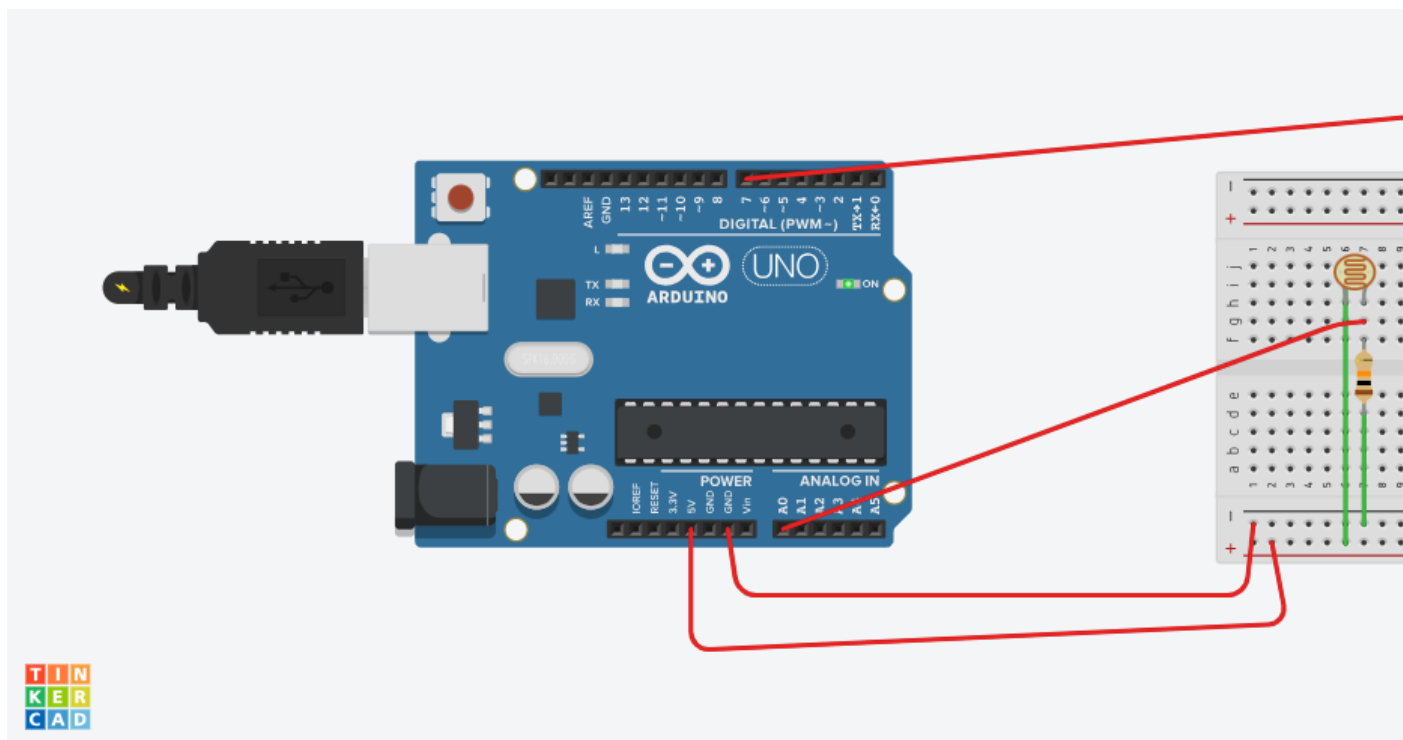
When a **light sensor (LDR)** is used, the sensor detects the surrounding light intensity. If the environment becomes dark, the microcontroller turns the **LED ON** automatically. When there is enough light, the microcontroller turns the **LED OFF**.

When a **motion sensor (PIR)** is used, the sensor detects the presence of a person. If motion is detected, the LED turns **ON**. If no motion is detected for a certain period, the LED turns **OFF**, helping to reduce unnecessary power consumption

WORKING PRINCIPAL

The Smart Lighting System works by using a sensor to monitor the surrounding environment and control an LED automatically. A microcontroller, such as an Arduino, continuously reads the data from the sensor. If a **Light Dependent Resistor (LDR)** is used, it detects the intensity of light in the surroundings. When the light level becomes low (dark), the microcontroller turns the LED **ON**. When sufficient light is available, it turns the LED **OFF**. If a **PIR motion sensor** is used instead, the sensor detects the movement of a person. When motion is detected, the microcontroller switches the LED **ON**, and when no motion is detected for a specified time, it switches the LED **OFF**. This automatic operation helps save electricity, reduces human effort, and makes the lighting system more efficient and convenient for homes, offices, and public places

CIRCUIT DIGARAM -



CODE-

```
// Smart Lighting System
```

```
// LED turns ON in darkness and OFF in daylight
```

```
const int sensorPin = A0; // LDR connected to Analog Pin A0
```

```
const int ledPin = D7; // LED connected to Digital Pin D7
```

```
int sensorValue = 0;

int threshold = 500;    // Adjust according to your room lighting

void setup()
{
    pinMode(ledPin, OUTPUT);
    Serial.begin(9600);
}

void loop()
{
    // Read LDR value
    sensorValue = analogRead(sensorPin);

    // Display sensor value on Serial Monitor
    Serial.print("LDR Value: ");
    Serial.println(sensorValue);

    // If it is dark, turn LED ON
    if(sensorValue < threshold)
    {
        digitalWrite(ledPin, HIGH);
    }
    else
    {
        digitalWrite(ledPin, LOW);
    }

    delay(200);
}
```

COMPONENTS –

- Arduino Uno (or any microcontroller)
- LED
- Light Sensor (LDR) or Motion Sensor (PIR)
- Resistor (220 Ω for LED, 10 k Ω for LDR if required)
- Breadboard and connecting wires
- Power supply

REAL WORLD APPLICATION –

- Automatic street lights
- Smart homes
- Office buildings
- Parking areas
- Garden and security lighting